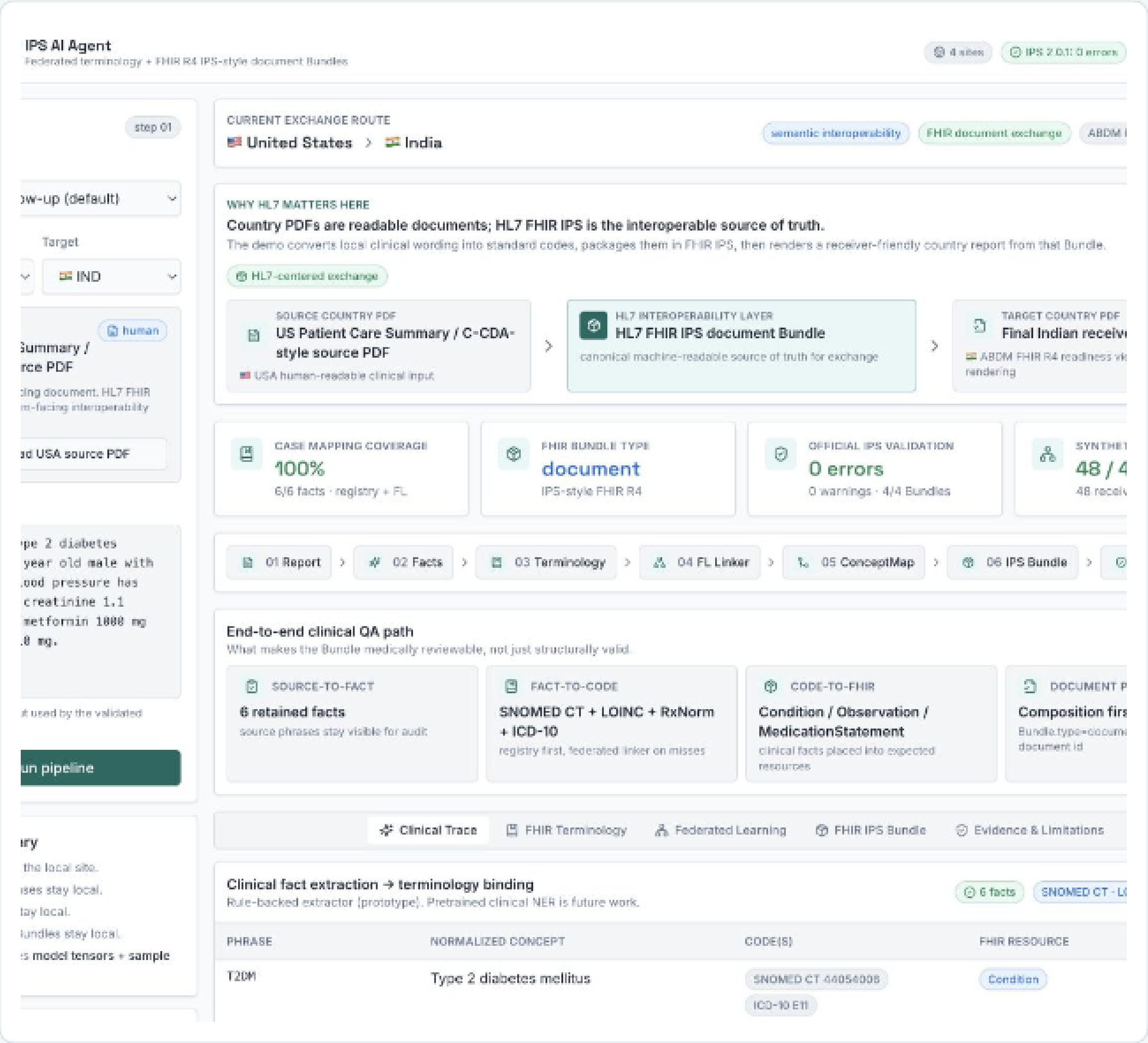
HL7 AI CHALLENGE 2026

Cross-Border IPS AI Agent

Federated, FHIR-native terminology alignment for cross-border Type 2 diabetes summaries.

FHIR IPS is the interoperable artifact.
PDFs are human-readable renderings.



Working demo · synthetic evidence · official IPS 2.0.1 validation

One condition arrives under many local names

USA	T2DM
India	sugar disease · madhumeha type 2
Australia	type 2 DM
Europe	DM2 · diabetes mellitus type II

→

Canonical meaning

Type 2 diabetes mellitus

SNOMED CT 44054006

ICD-10 E11

Without semantic alignment, a receiver EHR gets ambiguous text instead of computable clinical data.

FHIR IPS is the artifact that crosses the border

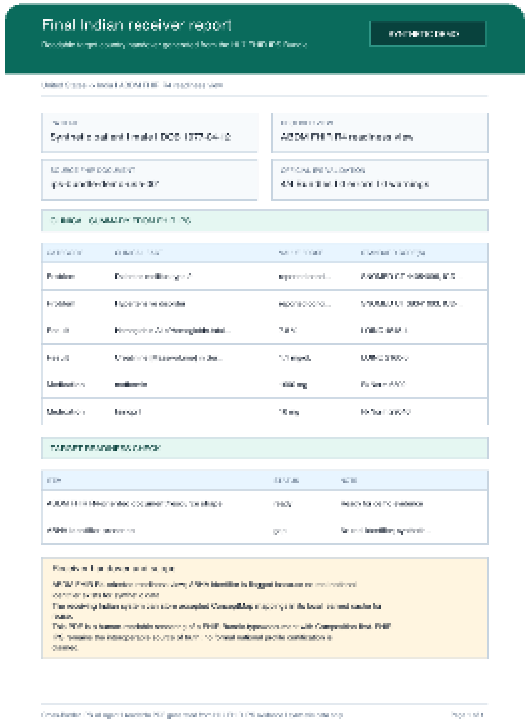
Human input

→

HL7 FHIR IPS

Bundle.type = document
Composition first
Coded clinical resources

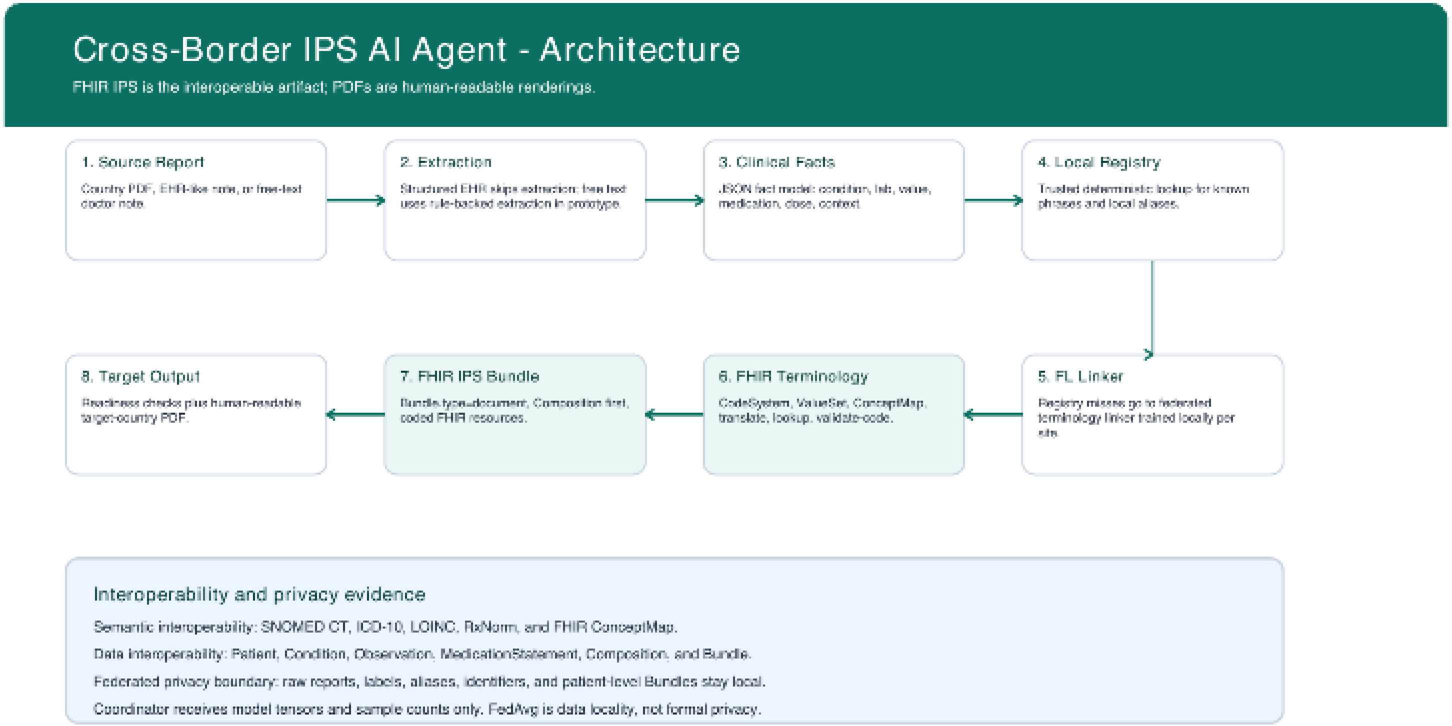
→



Human receiver view

Every transformation remains inspectable

The registry is deterministic; the federated linker is used only on misses; FHIR terminology validates and publishes the mapping.



Federation adds one cross-site mapping without moving reports

48/48

federated transfer

47/48

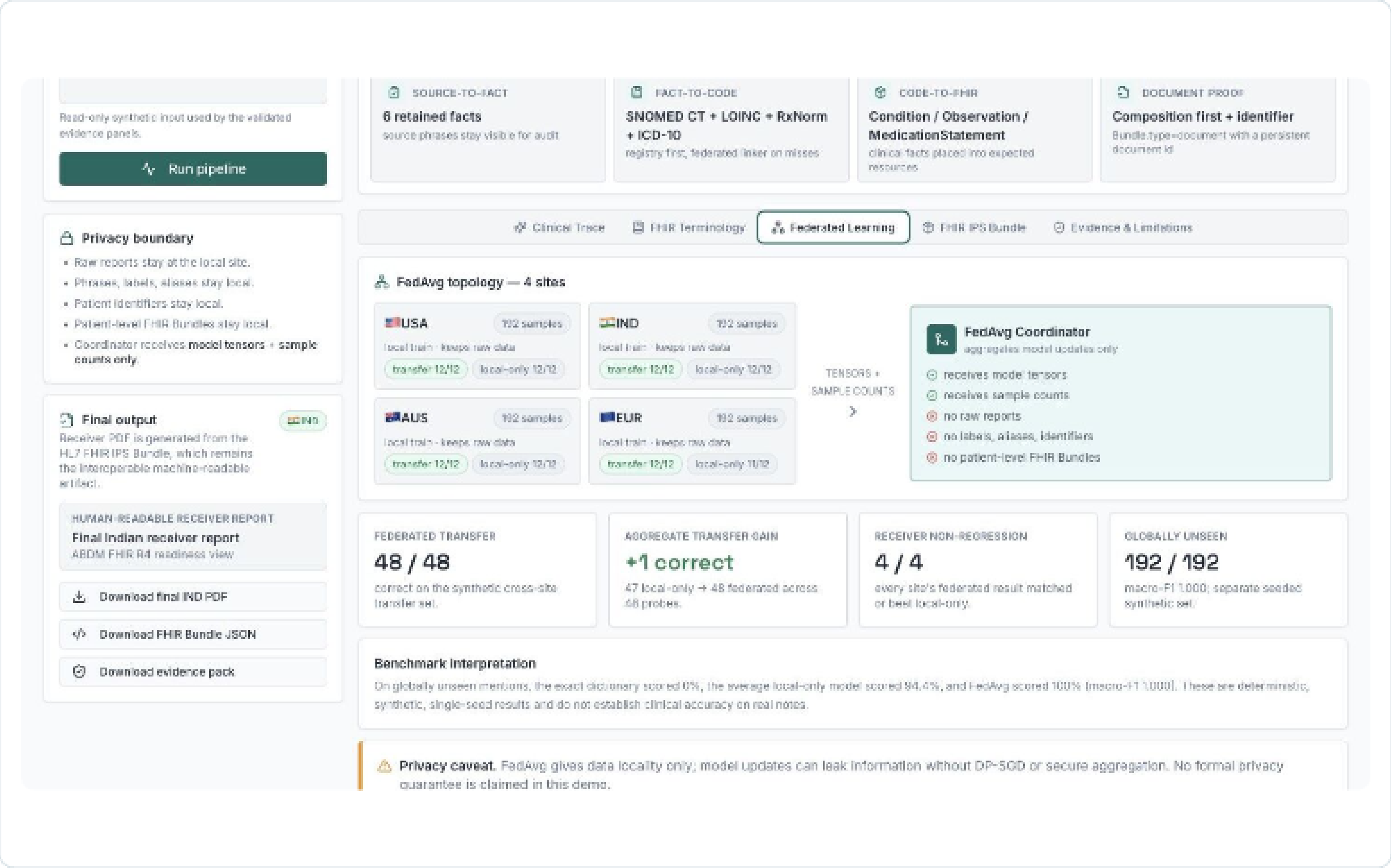
local-only transfer

+1

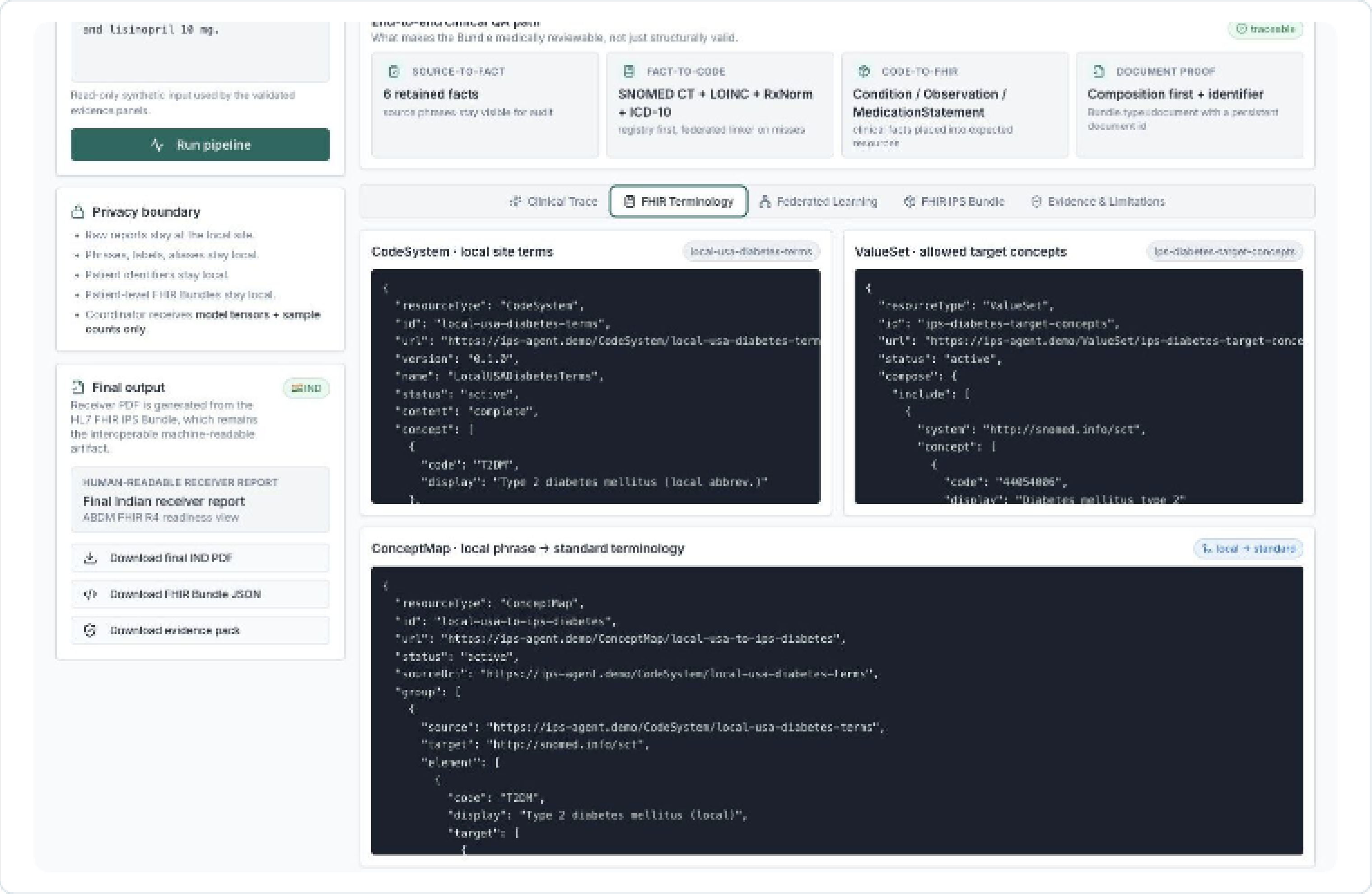
measured transfer gain

4/4 receiver sites did not regress

Single-seed synthetic benchmark; not clinical accuracy.



FHIR turns an AI prediction into an auditable mapping



1 **CodeSystem**
represents local site terms

2 **ValueSet**
constrains allowed targets

3 **ConceptMap**
publishes local → standard

4 **$translate / $lookup / $validate-code**
records validation evidence

Below 0.70 confidence → unresolved + human review

All four exchange Bundles pass the IPS 2.0.1 profile

4/4

current route Bundles

0

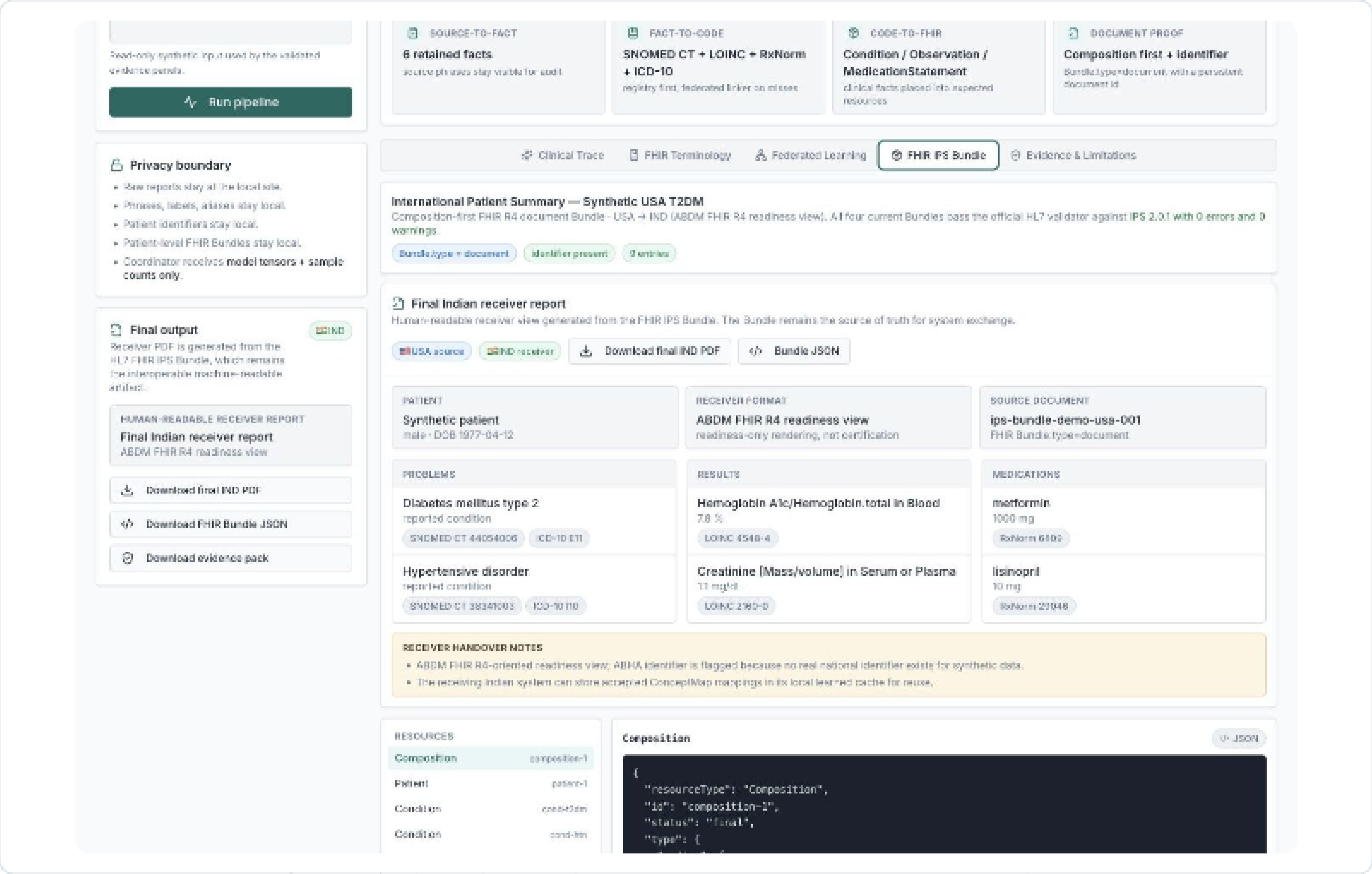
errors

0

warnings

2 information notes per Bundle

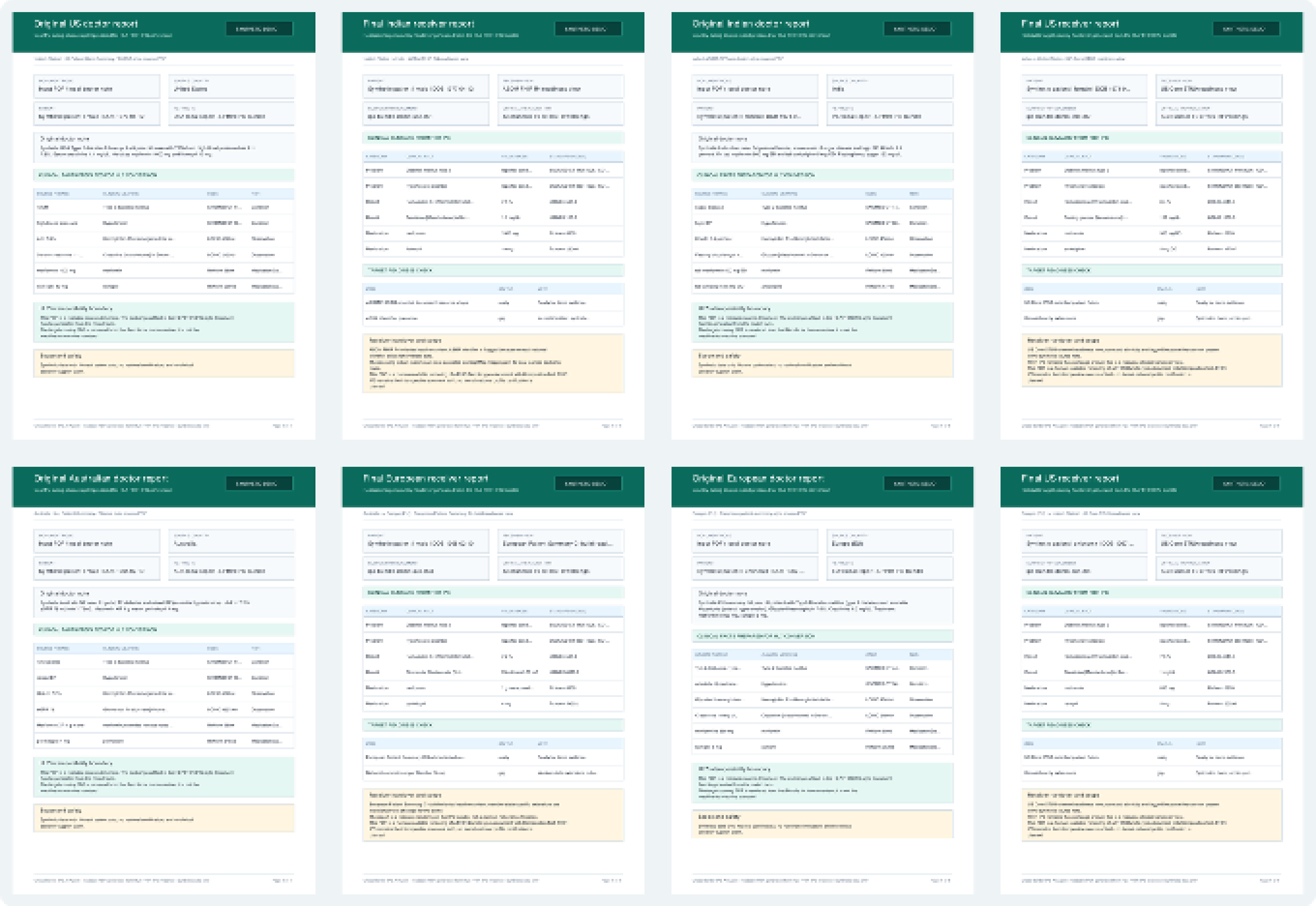
RxNorm ingredient codes are outside the IPS guide's recommended medication ValueSet.



Structural/profile conformance for synthetic Bundles; not clinical or national certification.

One machine-readable summary supports four receiver journeys

USA → India · India → USA · Australia → Europe · Europe → USA



The PDFs are readable renderings. The FHIR IPS Bundle remains the system-facing source of truth.

The evidence is executable, downloadable, and reviewable

15/15
frontend behavior and evidence tests

11/11
Python pipeline and FedAvg tests

16/16
route downloads click-tested

```
$ npm test -- --run
✓ 15 tests passed

$ PYTHONPATH=src pytest -q
✓ 11 tests passed

$ HL7 FHIR Validator + IPS 2.0.1
✓ 4 Bundles · 0 errors · 0 warnings
```

GitHub includes the real PyTorch clients, FedAvg coordinator, benchmark JSON, Bundles, PDFs, screenshots, and unedited validator logs.

THE CLOSING CLAIM

The standards do the traveling.
The reports stay local.

Federated learning

aligns local clinical terminology without centralizing raw reports

FHIR terminology

makes CodeSystem, ValueSet, ConceptMap, and validation evidence auditable

FHIR IPS

packages coded facts into a cross-border patient summary

Working demo

crossroad-fhir-link-three.vercel.app

Synthetic prototype · readiness checks only · no formal privacy guarantee